



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LAB 02 INSTRUCTIONS

For

Plan capture placement

Course Code: C1123

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.1

Lab 02 — Plan capture placement

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 02 — Capture Methods and Capture Filters

Learning outcome: LO1 — Plan scoped captures and configure a reproducible analyst profile.

Time: about 75 minutes

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Scenario

A user on the branch LAN reports slow web pages. Your manager asks where a sensor should go — and what it would see — before anyone touches the switch.

Goal

A switched access port normally observes its own traffic and broadcasts; SPAN or TAP placement changes visibility.

What you will produce

A completed capture plan with sensor placement and limits.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete
assets/topology.md	Fictional branch topology
assets/capture-plan.md	Capture plan template you complete

Lab at a glance

1. Mark three sensor points on the topology
2. Check capture-filter syntax (tcp port 80)
3. Compare capture vs display filtering
4. Complete the capture plan and permissions

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `assets/topology.md`. Draw a sensor on the client access port, on a mirrored uplink and beside the server. State which traffic each sees.

3. Open Capture > Options and inspect the interface list without starting a capture. Enter tcp port 80 in the capture-filter box and check syntax feedback. Cancel the dialog.

4. Open `data/branch-office.pcap`. Enter `tcp.port == 80` in the DISPLAY toolbar. Compare the stored total with the displayed total; explain why clearing it restores hidden packets.

5. Complete `assets/capture-plan.md` with scope, interface, duration, ring-buffer limit and permission owner. Wireless monitor mode requires compatible adapter/driver; USB capture requires platform-specific support. Remote capture should use authorised SSH/extcap, not obsolete unauthenticated RPC instructions.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

tcp.port == 80					
No.	Time	Source	Destination	Protocol	Info
18	1.040000	192.0.2.10	192.0.2.20	TCP	51001 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 W...
19	1.070000	192.0.2.20	192.0.2.10	TCP	80 → 51001 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0...
20	1.080000	192.0.2.10	192.0.2.20	TCP	51001 → 80 [ACK] Seq=1 Ack=1 Win=1048576 Len=0
21	1.090000	192.0.2.10	192.0.2.20	HTTP	GET /health HTTP/1.1
22	1.100000	192.0.2.20	192.0.2.10	TCP	80 → 51001 [ACK] Seq=1 Ack=71 Win=1048576 Len=0
23	1.140000	192.0.2.20	192.0.2.10	HTTP	HTTP/1.1 200 OK (text/plain)
24	1.150000	192.0.2.10	192.0.2.20	TCP	51001 → 80 [ACK] Seq=71 Ack=71 Win=8222720 Len=0
25	1.160000	192.0.2.20	192.0.2.10	TCP	80 → 51001 [FIN, ACK] Seq=71 Ack=71 Win=1048576 L...
26	1.170000	192.0.2.10	192.0.2.20	TCP	51001 → 80 [ACK] Seq=71 Ack=72 Win=1048576 Len=0
27	1.180000	192.0.2.10	192.0.2.20	TCP	51002 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 W...
... 27 more matching packets					
branch-office.pcap Displayed: 37					

Lab 02 expected evidence

Test it

The supplied `tcp.port == 80` expression matches 37 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching lab-tls.keys file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -D
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

Getting Started lab — list your interfaces and identify which one carries traffic before planning a capture. Source: J.F. Kurose and K.W. Ross, Wireshark Labs (gaia.cs.umass.edu/kurose_ross/wireshark.php).

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.